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Distributionally robust optimization for peer-to-peer energy trading considering data-driven ambiguity sets

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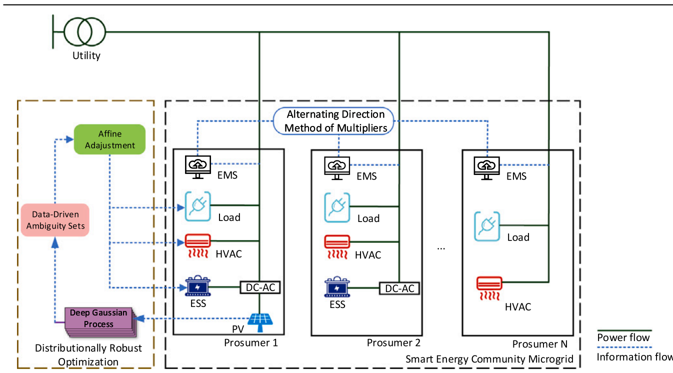
GRAPHICAL ABSTRACT



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ABSTRACT

Peer-to-peer (P2P) energy trading provides potential economic benefits to prosumers. The prosumers are responsible for managing their own resources/reserves within the energy community, especially for photovoltaic (PV). However, the intermittency of PV leaves a major issue for the optimal operation of P2P energy trading. This paper proposes a fully data-driven distributionally robust optimization (DRO) for P2P energy trading. Specifically, both the optimization approach and the ambiguity set of DRO are formed in a data-driven fashion. The proposed formulation minimizes the expected operation cost of each prosumer, which is modeled as a DRO problem considering the operational constraints. A decentralized energy negotiation mechanism and market clearing algorithm are proposed for P2P energy trading based on the alternating direction multiplier method. Furthermore, the ambiguity set is formed by deep Gaussian process under the framework of bootstrap aggregating. Finally, the equivalent linear programming reformulations of the proposed DRO model are carried out and solved in a distributed manner. Numerical results demonstrate that the proposed DRO-based approach has superior performance for handling the randomness of PV generation compared with robust optimization, stochastic programming, and other DRO variants.

1. Introduction

Energy equity, energy security, and environmental sustainability are regarded as "Energy Trilemma" [1], which is one of the core issues in modern society. An unprecedented global energy revolution is a challenging and ongoing topic for all governments. Although there is no "silver bullet" for breaking the dilemma, some potential approaches

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have been proposed, like interconnected energy systems and localized energy systems [2]. With the advent of distributed energy resources (DERs) and advanced information and communication technology (ICT), traditional consumers can generate power for self-consumption or trading with the utility. The deregulated electricity market has been a consensus and the prosumer era has encountered in recent years [3]. Thus, Peer-to-Peer (P2P) energy trading, considered the next-generation local electricity market paradigm, has attracted a lot of research attention [4].

Since the implementation of P2P energy trading is beneficial to renewable energy consumption, peak shaving, and economic prospects, a number of studies have been proposed on P2P energy trading. Generally, the majority of P2P energy trading mechanisms include game theory (GT), auction theory, and constrained optimization [5]. For the GT-based P2P energy trading, a Stackelberg game approach is proposed for P2P energy trading in virtual microgrids [6]. In [7], a stochastic game scheme is introduced for P2P energy sharing considering the social attributes. Ref. [8] gives an electricity trading approach based on a coalition graph game. Regarding the auction theory, a prediction-integrated continuous double auction model is proposed in [9,10]. In [11], an auction-based strategy is designed for intraday P2P energy trading. Furthermore, a blockchain-enabled double auction approach also gives some successful cases for P2P energy trading [12]. It should be noted that the auction-based approach can be formed in a GT manner [13]. The constrained optimization approach has been heavily explored to design the P2P energy trading by linear programming (LP) [14], mixed-integer linear programming (MILP) [15], and consensus-based approach [16]. Furthermore, the reinforcement learning-based approaches [17,18] are also considered to belong to constrained optimization, where rewards and policies can be regarded as special variants of objective functions and optimal solutions.

Throughout the aforementioned P2P energy trading mechanism, the randomness of renewable power generation is ignored in the majority study. The randomness of renewable power generation will break the pre-defined energy dispatch schemes and even fall into infeasible solutions. Therefore, it is of great significance to involve renewable power generation randomness in P2P energy trading, especially for photovoltaics (PV). Some prior works including the robust optimization (RO) [19,20], stochastic programming (SP) [21,22], and distributionally robust optimization (DRO) [23] models, have been proposed to handle the randomness of PV generation. However, the RO-based approach merely concentrates on the worst-case scenarios, which leads to an over-conservative optimal solution. The sampling average approximation (SAA) [24] is a good approach for measuring the expected value of SP, but an exact distribution of PV generation has to be pre-defined and any deviants may cause a sub-optimal solution. Furthermore, the “curse of dimensionality” is often encountered when dealing with multi-scenarios/stages problems. The state-of-art approach for evaluating the expectation value is a DRO-based approach [25]. The DRO is a data-driven optimization approach, where data is sampled from historical data (called ambiguity set). The ambiguity set is constructed by the statistics of historical data, like empirical probability density function [26], Wasserstein metric [27]. However, these approaches do not consider the correlation between the attribute and generation, only concentrating on the maximal and minimal forecast error from historical data. Although using historical forecast error is simple and effective, it will lead to a conservative solution in many special cases.

Over the last decade, explosive development has been witnessed for artificial intelligence (AI) and its application. The AI-based approaches have a natural capacity to construct the interval of forecasted value and prior work has been proposed like Bayesian Deep Learning (BDL) [28], and Gaussian Process (GP) Regression [29]. However, to our knowledge, there is a research gap to cooperate with an AI-based approach to resolve P2P energy trading in the framework of DRO. Thus, this paper endeavors to fill this gap by proposing a fully data-driven approach to constructing an optimization model for P2P energy trading. Specifically, both the optimization approach and the ambiguity set are constructed in a data-driven manner. Firstly, a DRO model is proposed for each prosumer to minimize operating costs. Then, a decentralized energy negotiation strategy and market clearing algorithm are designed for P2P energy trading. Finally, the ambiguity set is formed by employing a novel deep GP (DGP) algorithm and the tailor-made reformulated LP problem is derived. The main contributions are summarized as follows:

1. A novel DGP variant, i.e., implicit posterior variational inference DGP, is utilized to quantify the uncertainties of PV generation. The existing point forecast models pose a big challenge to the operators with uncertainty-aware decisions. Specifically, the predefined energy dispatch schedules will fall into the sub-optimal or infeasible. The proposed DGP can achieve inductive reasoning effectively and enhance its ability in quantifying the uncertainties of PV generation.
2. A novel data-driven ambiguity set is proposed for modeling the P2P energy trading based on implicit posterior variational inference DGP in the bootstrap aggregating framework. The proposed ambiguity set is more precise than most existing statistical-based ambiguity sets. The tailor-made equivalent LP model is derived based on the proposed ambiguity set. To our best knowledge, this is the first attempt to integrate the DRO model and machine learning (ML) approach to P2P energy trading. The proposed data-driven ambiguity sets enabled the DRO model to minimize the operating cost by coordinating local energy management while encountering the randomness of PV generation.

Numerical results of the proposed DRO compared with other DRO variants have been carried out. It shows that our proposed approach has superior performance on the economic benefits and computational feasibility for P2P energy trading.

The remainder of this paper is organized as follows. Section 2 describes the mathematical formulation of P2P energy trading. Section 3 introduces a decentralized energy negotiation mechanism. The problem is reformulated based on the data-driven ambiguity set in Section 4. The numerical results are presented in Section 5. Finally, Section 6 concludes this work.

2. Mathematical formulation

2.1. General framework of a smart energy community

The structure of the smart energy community microgrid for P2P energy trading between prosumers is shown in Fig. 1(a). Each prosumer has its own Heating, Ventilating and Air Conditioning (HVAC), and basic load, with some prosumers installing PV panels and energy storage systems (ESS) as well. The PV and ESS are connected to the household load by a DC–AC converter. All the prosumers are connected with each other by bidirectional power lines and information exchange links. The power line for the whole energy community is connected to the upstream utility. Local energy management systems (EMS) that integrate P2P market clearing algorithms are equipped for each prosumer. The EMS can monitor the generation, consumption, and energy trading with other prosumers and/or the utility. The general framework of the proposed DRO model is shown in Fig. 1(b). The prosumers can schedule their energy usage, and decide to trade with the utility and/or other peers. Furthermore, since the generation of PV is uncertain when doing the day-ahead scheduling, the implicit posterior variational inference DGP is utilized to quantify the uncertainties. Then, the data-driven ambiguity set is formed by the Bootstrap Aggregating (Bagging) algorithm. Finally, the proposed DRO model is adopted to minimize the overall cost of the prosumer and provide an affine adjustment factor. The affine adjustment factor is used to re-schedule their energy usage during the real-time stage.

Based on the existing infrastructure within the smart energy community, this study aims to propose a P2P energy trading strategy that is sufficient for handling the randomness of PV. Let $\mathcal{I} = \{1, 2, \ldots, I\}$ be the set of prosumers in the energy community. The neighboring set of prosumer i is denoted as N_i, where $n_i \triangleq \|N_i\|_1$. For each prosumer, $\vec{p}_i = (p_{i1}, p_{i2}, \ldots, p_{in_i}) \in \mathbb{R}^{n_i}$ is the vector of traded power between peers i and j, $1 \leq j \leq n_i$. p_i is total traded power within prosumer i, where $p_i = \sum_{j=1}^{n_i} p_{ij}$.

2.2. Energy storage system

It is supposed that the ESS is equipped for each prosumer. The dynamic process of charge and discharge is modeled as shown in (1).

$$0 \leq c_{i,h} \leq \overline{c}_i \qquad 0 \leq d_{i,h} \leq \overline{d}_i \tag{1a}$$

$$S_{i,h} = (1 - \eta^{loss}) S_{i,h-1} + \eta^c c_{i,h} \Delta h - \eta^d d_{i,h} \Delta h \tag{1b}$$

$$\underline{S}_i \leq S_{i,h} \leq \overline{S}_i \tag{1c}$$

$$S_{i,H} \geq S_{i,0} \tag{1d}$$

where $c_{i,h}$ and $d_{i,h}$ are the charge power and discharge power of the ESS for each prosumer during the time slot h. $\overline{c}_i$ and $\overline{d}_i$ are the maximum power limits for charge and discharge, respectively. The energy level is denoted as $S_{i,h}$ and is limited between $\underline{S}_i$ and $\overline{S}_i$. η^{loss} is energy leakage rate, which is subject to $\eta^{loss} \ll 1$. The charging and discharging efficiency are represented as η^c and η^d and $\eta^c \leq 1 \leq \eta^d$.

The operation cost of the degradation for ESS is estimated based on [30] and shown in (2).

$$C_i^{ESS} = \sum_h \rho_i (c_{i,h} + d_{i,h}) \Delta h \tag{2}$$

where ρ_i is the levelized cost of ESS within the lifetime.

2.3. Heating, ventilating, and air conditioning unit

The HVAC unit is an important electrical appliance and provides a comfortable environment within the buildings. The thermodynamic model of a HVAC unit [31] between the current indoor temperature ($T_{i,h}^{in}$), previous indoor temperature ($T_{i,h-1}^{in}$) and current outdoor temperature (T_h^{out}) is shown in (3).

$$T_{i,h}^{in} = T_{i,h-1}^{in} - \frac{T_{i,h-1}^{in} - T_h^{out} + \eta_i R_i q_{i,h}}{C_i R_i} \Delta h \tag{3}$$

where C_i and R_i are the thermal capacity and thermal resistance of the HVAC unit, respectively. η_i denotes the energy efficiency of HVAC operation. $q_{i,h}$ denotes energy consumption.

The indoor temperature needs to be kept within a suitable range to ensure it is habitable. Besides, the energy demand of the HVAC unit is also a continuously adjusted variable and is limited below the maximum power.

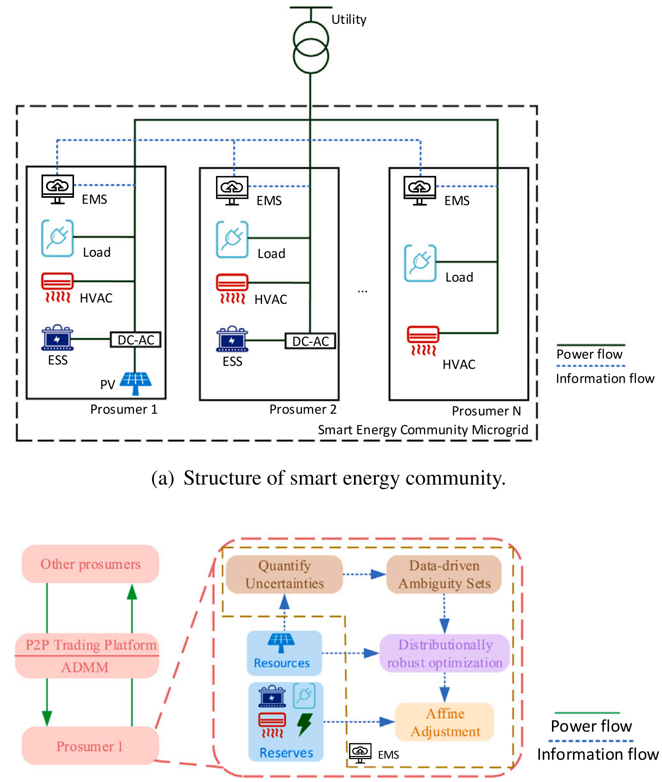
$$\underline{T}_i^{in} \leq T_{i,h}^{in} \leq \overline{T}_i^{in} \tag{4a}$$

$$0 \leq q_{i,h} \leq \overline{q}_i \tag{4b}$$

where $\underline{T}_i^{in}$ and $\overline{T}_i^{in}$ are the minimum and maximum acceptable indoor temperature, $\overline{q}_{i,h}$ is the maximum power of the HVAC unit for prosumer i. It should be noted that the indoor temperature range is considered as the virtual energy stored in the building. Virtual energy can participate in demand response (DR) for prosumers, the economic prospect for DR is discussed in Section 5.

2.4. Trading with the utility

The prosumer can purchase energy from the utility when the energy demand cannot be satisfied by P2P energy trading. Conversely, if no consumer wishes to purchase more energy, the redundant energy will be fed back to the utility. The energy trading cost with the utility is shown in (5).



(a) Structure of smart energy community.

(b) General framework of the proposed DRO model.

Fig. 1. Structure and framework for prosumers within the smart energy community.

$$C_i^{Utility} = \sum_h (p_h^b b_{i,h} - p_h^s s_{i,h}) \Delta h \tag{5}$$

where $b_{i,h}$ and $s_{i,h}$ denote the energy purchasing and selling profiles of the prosumer trading with the utility at p_h^b and p_h^s.

2.5. Exogenous cost

The exogenous cost of P2P energy trading is treated as the sum of network fee and implement cost [32]. In this paper, the allocation policy of exogenous cost is realized by electrical distance and traded energy volume between peers, the formula is shown in (6).

$$C_i^{P2P} = \sum_h \sum_{j \in N_i} \pm \frac{u^{dist} d_{ij}}{2} p_{ij} \Delta h \tag{6}$$

where d_{ij} is the electrical distance between peer i and peer j. u^{dist} is bilateral trade weight and expressed in \$/(kWh·distance unit). The signal $\pm$ is subject to the role in the P2P energy trading, that is, positive is for producers and negative for consumers. The exogenous costs will be transferred to the utility as compensation for network usage. The electrical distance shows the level of energy loss on the line, which is defined by Thevenin impedance distance [33] and shown in (7).

$$d_{ij} = |Z_{ii} + Z_{jj} - Z_{ij} - Z_{ji}| \tag{7}$$

where $\boldsymbol{Z}$ is branch impedance matrix within the microgrid, Z_{ii} and Z_{ij} are self-impedance and mutual impedance of bus i, respectively.

2.6. Affine adjustment for uncertainties

Given the randomness of PV generation, the re-scheduling of local energy/reserve is essential to keep the energy balance. Thus, the reserve is provided by the HVAC unit, ESS, and/or trading with the utility to reduce the deficit of energy usage. The mechanism of affine adjustment [34] is shown in (8)

$$q_{i,h} \Leftarrow q_{i,h} + (\alpha^{q,+}_{i,h} - \alpha^{q,-}_{i,h})\xi_{i,h} \quad 0 \le \alpha^{q,+}_{i,h}, \alpha^{q,-}_{i,h} \le 1 \tag{8a}$$

$$c_{i,h} \Leftarrow c_{i,h} + (\alpha^{c,+}_{i,h} - \alpha^{c,-}_{i,h})\xi_{i,h} \quad 0 \le \alpha^{c,+}_{i,h}, \alpha^{c,-}_{i,h} \le 1 \tag{8b}$$

$$d_{i,h} \Leftarrow d_{i,h} + (\alpha^{d,+}_{i,h} - \alpha^{d,-}_{i,h})\xi_{i,h} \quad 0 \le \alpha^{d,+}_{i,h}, \alpha^{d,-}_{i,h} \le 1 \tag{8c}$$

$$b_{i,h} \Leftarrow b_{i,h} + (\alpha^{b,+}_{i,h} - \alpha^{b,-}_{i,h})\xi_{i,h} \quad 0 \le \alpha^{b,+}_{i,h}, \alpha^{b,-}_{i,h} \le 1 \tag{8d}$$

$$s_{i,h} \Leftarrow s_{i,h} + (\alpha^{s,+}_{i,h} - \alpha^{s,-}_{i,h})\xi_{i,h} \quad 0 \le \alpha^{s,+}_{i,h}, \alpha^{s,-}_{i,h} \le 1 \tag{8e}$$

$$\sum_{k \in \{q,c,d,b,s\}} \alpha^{k,+}_{i,h} - \alpha^{k,-}_{i,h} = 1 \tag{8f}$$

where $\alpha^{q,+}_{i,h}$, $\alpha^{q,-}_{i,h}$, $\alpha^{c,+}_{i,h}$, $\alpha^{c,-}_{i,h}$, $\alpha^{d,+}_{i,h}$, $\alpha^{d,-}_{i,h}$, $\alpha^{b,+}_{i,h}$, $\alpha^{b,-}_{i,h}$ and $\alpha^{s,+}_{i,h}$, $\alpha^{s,-}_{i,h}$ are positive or negative affine participation factors for energy consumption of the HVAC units, charging/discharging operation, and energy buying/selling with utility, respectively. $\xi_{i,h}$ is the deviation of expected PV generation at timeslot h.

Remark 1. The randomness of PV generation will have an effect on P2P energy trading, but the volume of P2P energy trading is determined by a consensus-based algorithm. The affine adjustment to P2P energy trading will lead to non-converge when prosumers have different PV generation deviations.

2.7. Objective function

A DRO model is formed for P2P energy trading to mitigate the randomness of renewable energy generation. The overall cost consists of independent cost which is not affected by forecast deviation and dependent cost which is associated with forecast deviation. Thus, the objective of each prosumer is to minimize the degradation cost of ESS, trading cost with the utility, and exogenous cost of P2P energy trading. Provided that the forecast deviation (ξ_i) follows a probability distribution, the worst-case cost of each prosumer is shown in (9).

$$\min \ \{C^{P2P}_i + \sup_{\mathbb{P}_i \in \mathcal{P}_i} \mathbb{E}_{\mathbb{P}_i}[C_i(\boldsymbol{x}_i, \xi_i)]\} \tag{9a}$$

$$s.t. \quad C_i = C^{ESS}_i + C^{Utility}_i \tag{9b}$$

$$p_{ij,h} + p_{ji,h} = 0 \tag{9c}$$

$$g_{i,h} + d_{i,h} + b_{i,h} + p_{i,h} \ge l_{i,h} + q_{i,h} + c_{i,h} + s_{i,h} \tag{9d}$$

$$\underline{f}_i \le b_{i,h} - s_{i,h} + p_{i,h} \le \overline{f}_i \tag{9e}$$

$$(1)\text{–}(8) \tag{9f}$$

$$\boldsymbol{x}_i := \{q_{i,h}, \alpha^q_{i,h}, c_{i,h}, \alpha^c_{i,h}, d_{i,h}, \alpha^d_{i,h}, b_{i,h}, \alpha^b_{i,h}, s_{i,h}, \alpha^s_{i,h}\}$$

where $\mathcal{P}_i$ is all possible probability distributions of PV generation and $\mathbb{P}_i$ is the worst probability distribution among all scenarios. $g_{i,h}$ and $l_{i,h}$ represent the solar power generation and the load of prosumer i at the timeslot h. The $\underline{f}_i$ and $\overline{f}_i$ are maximum power import/export of prosumer i. (9)c is a coupled constraint between peers in P2P energy trading. (9)d represents the local energy balance for each prosumer, it is worth pointing out that the wise prosumer will not buy and sell energy simultaneously since it is naturally mutually exclusive in the objective function. (9)e denotes the power import/export within prosumer i.

Remark 2. For the physical viability of P2P energy trading, we believe that it is the responsibility of the distribution system operator (DSO). Some techniques, like distribution network reconfiguration or reactive compensation, can be adopted by DSO to realize the optimal power flow. The operations cost can be compensated by the exogenous cost of P2P energy trading.

3. Decentralized energy negotiation mechanism

3.1. Energy negotiation mechanism

The energy negotiation between peers is a fully deregulated, bilateral, and decentralized process, which is achieved by the Alternating Direction Method of Multipliers (ADMM) [35]. The aim of energy negotiation is to minimize the overall operation cost by solving the local optimization problem and information exchange. Multiple prosumers are involved in P2P energy trading, but prosumers do not want to share their equipment parameters and energy usage, especially for commercial customers. Therefore, the existing n-block ADMM which updates variables parallel and shares limited information can be used [36].

To implement the decentralized market clearing algorithm, auxiliary variables are introduced to reformulate (9)c and shown as follows.

$$y_{ij,h} + y_{ji,h} = 0 \tag{10a}$$

$$y_{ij,h} = p_{ij,h} \tag{10b}$$

where $y_{ij,h}$ and $y_{ji,h}$ are the auxiliary variables.

3.2. Decentralized market clearing algorithm

The dual variable ($\boldsymbol{\pi}_{ij}$) associated with (9)c is considered as the price of P2P energy trading, which is also called shadow (dual) price in the economic perspective. The augmented Lagrangian can be formulated as follows.

$$\mathcal{L}_i(\boldsymbol{x}_i, \boldsymbol{y}_i, \boldsymbol{\pi}_i) = \sup_{\mathbb{P}_h \in \mathcal{P}_h} \mathbb{E}_{\mathbb{P}_h} C_i(\boldsymbol{x}_i, \xi_i) + \sum_{i \in \mathcal{I} \setminus i} [\pi_{ij,h}(p_{ij,h} - y_{ij,h}) + \frac{\psi}{2} \|p_{ij,h} - y_{ij,h}\|^2_2] \tag{11}$$

where ψ is the parameter for ADMM.

The decentralized market clearing algorithm is resolved by the following steps.

3.2.1. Update of $\boldsymbol{x}_i$

Due to the decomposability of augmented Lagrangian and the constraints (9)b and (9)d–(9)f, $\boldsymbol{x}_i$ is a fully local problem and is able to updating in a fully decentralized manner. Each prosumer i solves the problem (12) to updating its local variable $\boldsymbol{x}^{\tau}_i$ by pre-defined the auxiliary variable and dual variable as follows.

$$\boldsymbol{x}^{\tau+1}_i := \arg\min_{\boldsymbol{x}_i} \mathcal{L}_i(\boldsymbol{x}_i, \boldsymbol{y}^{\tau}_i, \boldsymbol{\pi}^{\tau}_i) \tag{12a}$$

$$s.t. \quad (9)\text{b}, (9)\text{d–}(9)\text{f} \tag{12b}$$

where τ is the iteration index.

3.2.2. Update of $\boldsymbol{y}_i$

The auxiliary variable $\boldsymbol{y}_i$ is updated by solving the problem (13) and shown as follows.

$$y^{\tau+1}_{ij,h}, y^{\tau+1}_{ji,h} := \arg\min_{y_{ij,h}, y_{ji,h}} -\pi^{\tau}_{ij,h} y_{ij,h} - \pi^{\tau}_{ji,h} y_{ji,h} + \frac{\psi}{2}[(p^{\tau+1}_{ij,h} - y_{ij,h})^2 + (p^{\tau+1}_{ji,h} - y_{ji,h})^2] \tag{13a}$$

$$s.t. \quad (10) \tag{13b}$$

3.2.3. Update of $\boldsymbol{\pi}_{ij}$

The dual variable is updated by standard ADMM and shown in (14).

$$\pi^{\tau+1}_{ij} = \pi^{\tau}_{ij} + \psi(p^{\tau+1}_{ij,h} - y^{\tau+1}_{ij,h}) \tag{14}$$

3.2.4. Convergence criteria

The convergence criteria of the iterative process are estimated based on the primal and dual residuals, which is shown in (15).

$$\|r^{\tau}\|_2 \le \epsilon^{pri} \qquad \|s^{\tau}\|_2 \le \epsilon^{dual} \tag{15}$$

where $r^{\tau} \triangleq p^{\tau} - y^{\tau}$ and $s^{\tau} \triangleq -\psi(y^{\tau} - y^{\tau-1})$. The primal and dual residuals feasibility tolerances are chosen by [35].

$$\epsilon^{pri} = \sqrt{m}\epsilon^{abs} + \epsilon^{rel}\max\{\|p^{\tau}\|_2, \|-y^{\tau}\|_2\} \tag{16a}$$

$$\epsilon^{dual} = \sqrt{m}\epsilon^{abs} + \epsilon^{rel}\|\pi^{\tau}\|_2 \tag{16b}$$

Here, the ϵ^{abs} and ϵ^{rel} are absolute tolerances and relative tolerances and set as 10^{-4} and 10^{-2}, respectively. m is the number of linear constraints (10)a.

4. Reformulation based on the data-driven ambiguity set

In this section, a DGP is introduced to quantify the randomness of PV generation. Then, the data-driven ambiguity set is constructed based on DGP under the framework of bagging. Finally, the problem reformulation is formed based on the ambiguity set.

4.1. Quantitative PV generation randomness

For quantifying the randomness of PV generation, a probabilistic forecasting approach is proposed using the implicit posterior variational inference DGP [37]. The GP is a Bayesian non-parametric method assuming that a collection of attributes are subject to a joint Gaussian distribution. Since the time complexity of GP is $\mathcal{O}(N^3)$ (N is the number of data points), which is a big concern in the big-data era. To mitigate the concern of time cost, the sparse GP [38] is developed by introducing M inducing points and the time complexity is $\mathcal{O}(NM^2)$ ($M < N$). However, the GP and its variants rely on a sophisticated covariance function and it is questionable in the application of the datasets with different similarity metrics in different regions of the input space. The DGP overcomes this limitation by using a stacked GP structure, which has a natural capacity for the non-stationary data [39]. This article further extends the sparse GP with a stacked structure and the implicit posterior variational inference generates the PV generation with a more general distribution type.

Assume N training samples $\mathcal{X} \triangleq \{\mathfrak{x}_n\}_{n=1}^{N}$ and their observations $\mathcal{Y} \triangleq \{\mathfrak{y}_n\}_{n=1}^{N}$. For a DGP with the depth being L, each layer has a set of input ($\mathfrak{f}_{l-1}$) and output ($\mathfrak{f}_l$), where $\mathcal{F} \triangleq \{\mathfrak{f}_l\}_{l=1}^{L}$. Consider a stacked sparse GP with inducing inputs $\mathcal{Z} \triangleq \{\mathfrak{z}_l\}_{l=1}^{L-1}$ and corresponding output variables $\mathcal{U} \triangleq \{\mathfrak{u}_l\}_{l=1}^{L}$, respectively. The joint probability distribution of the sparse GP is as follows.

$$p(\mathcal{Y}, \mathcal{F}, \mathcal{U}) = \prod_{n=1}^{N} p(\mathfrak{y}_n|\mathfrak{f}_i^L) \prod_{l=1}^{L} p(\mathfrak{f}^l|\mathfrak{u}^l; \mathfrak{f}^{l-1}, \mathfrak{z}^{l-1})p(\mathfrak{u}^l; \mathfrak{z}^{l-1}) \tag{17}$$

where $\mathfrak{f}^0 \triangleq \mathcal{X}$. In the DGP model, the marginal likelihood is obtained by the variational inference. The inferred posterior is shown in (18).

$$q(\mathcal{F}, \mathcal{U}) = \prod_{n=1}^{L} p(\mathfrak{f}^l|\mathfrak{u}^l; \mathfrak{f}^{l-1}, \mathfrak{z}^{l-1})q(\mathfrak{u}^l) \tag{18}$$

where $q_\phi(\mathcal{U}) \triangleq \int p(\mathcal{U}|\epsilon)d\epsilon$ is parameterized by ϕ and a random noise $\epsilon \sim \mathcal{N}(0,1)$.

Considering the variational posterior ($q_\phi(\mathcal{U})$) implicitly makes it impossible to evaluate the evidence lower bound (ELBO), thus an adversarial-based inference approach with generator and discriminator is proposed. Observed that the KL divergence in the ELBO is equal to the expectation of the log-density ratio, the discriminator is denoted as a maximization problem, and the formula is shown in (19).

$$\max_{\kappa}\ \mathbb{E}_{p(\mathcal{U})}[\log(1-\sigma(T_\kappa(\mathcal{U})))] + \mathbb{E}_{q_\phi(\mathcal{U})}[\log(\sigma(T_\kappa(\mathcal{U})))] \tag{19}$$

where the $\sigma(\cdot)$ is a Sigmoid function, and $T_\kappa(\mathcal{U}) \triangleq \sum_{l=1}^{L} T_{\kappa_l}(u_l)$ is discriminator and parameterized by κ.

For the generator, the ELBO is formed using the estimated optimal discriminator. The optimization goal of the generator is shown in (20).

$$\max_{\theta,\phi}\ \mathbb{E}_{q_\phi(\mathcal{U})}[\mathcal{J}(\theta, \mathcal{X}, \mathcal{Y}, \mathcal{U}) - T_\kappa(\mathcal{U})] \tag{20}$$

where $\mathcal{J}(\theta, \mathcal{X}, \mathcal{Y}, \mathcal{U}) \triangleq \mathbb{E}_{p(\mathfrak{f}_L|\mathcal{U})}[\log p(\mathcal{Y}|\mathfrak{f}_L)]$ and θ denotes the hyperparameters of DGP.

ELBO optimization is considered as a pure-strategy game between the discriminator and generator. The optimal κ^*, ϕ^*, and θ^* are obtained via gradient ascent until a Nash equilibrium of the game is reached.

The quality of the forecast result is evaluated based on three error indices, i.e., interval coverage probability (ICP), Pinball loss and coefficient of determination (R^2), which are defined in (21).

$$\text{ICP} = \frac{1}{N}\sum_{i=1}^{N} \mathfrak{b}_i \tag{21a}$$

$$\text{Pinball} = \begin{cases} \frac{1}{NQ}\sum_{q=1}^{Q}\sum_{i=1}^{N}(g_i - \tilde{g}_{i,q})q, \tilde{g}_{i,q} < g_i \\ \frac{1}{NQ}\sum_{q=1}^{Q}\sum_{i=1}^{N}(\tilde{g}_{i,q} - g_i)(1-q), \tilde{g}_{i,q} > g_i \end{cases} \tag{21b}$$

$$R^2 = 1 - \frac{\sum_i (g_i - \tilde{g}_i)^2}{\sum_i (g_i - \bar{g})^2} \tag{21c}$$

where $\mathfrak{b}_i$ is a bool value, which indicates whether the forecast result is within the forecast interval, $\tilde{g}_{i,q}$ represents the load forecasting value at the qth quantile, and $\bar{g}$ is the mean value of PV generation. It should be noted that the ICP is used to assess the quality of the forecast interval, the next term in (21)b is to evaluate the forecasted probability, while the last index in (21)c is to assess the quality of forecast point.

4.2. Data-driven ambiguity set

For fixing the size of the ambiguity set, an ensemble learning framework is proposed based on the Bagging algorithm [40]. The size of the ambiguity set is determined by the combination of weak learners within the ensemble learning framework. Specifically, the upper bound and lower bound of the ambiguity set are selected as the maximum probability and the minimum probability among the weak learners, respectively. For mathematical conciseness, the superscripts h will be dropped in the following. The ambiguity set of prosumer i is as follows.

$$\mathcal{G}_i = \{\mathbb{P}_i[\tilde{g}_i \le \tilde{g}_i^k] \in [\underline{\mathfrak{p}}_i^k, \overline{\mathfrak{p}}_i^k] |\ \mathbb{P}_i \in \mathcal{P}_i[\underline{g}_i, \overline{g}_i]\} \tag{22}$$

where $\mathbb{P}_i$ is a probability measure/distribution for the PV generation, $\mathcal{P}_i[\underline{g}_i, \overline{g}_i]$ denotes the set of all probability measures whose supports (Ξ_i) are the interval $[\underline{g}_i, \overline{g}_i]$, $\overline{\mathfrak{p}}_i$ and $\underline{\mathfrak{p}}_i$ are the minimal and maximum probability within the support, and k is the number of sampling points within the support.

To solve the long-tail characteristic of Gaussian distribution, two Dirac distributions are proposed at both ends of the support set, which make the integral within the support is equal to 1. The Dirac distribution are shown in (23).

$$\underline{\mathbb{P}}_i = \underline{\mathfrak{p}}_i * \delta(g_i - \underline{g}_i) \tag{23a}$$

$$\overline{\mathbb{P}}_i = \overline{\mathfrak{p}}_i * \delta(g_i - \overline{g}_i) \tag{23b}$$

where $\delta(g_i)$ is a classical Dirac distribution.

The scheme is shown in Fig. 2 and the weak learner is considered as DGP in this paper, which is called bagging DGP in the following for convenience. The bagging scheme can improve the ability of randomness quantification and provide a more precise ambiguity set.

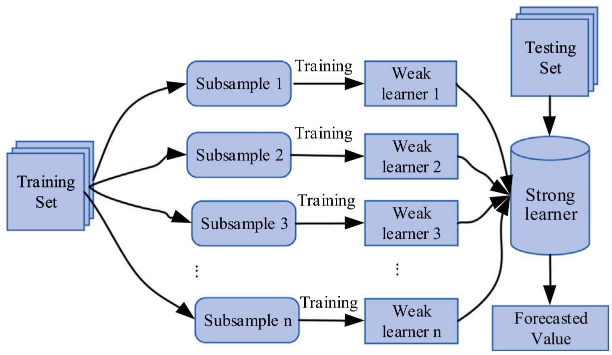


Fig. 2. The scheme of bagging ensemble learning.

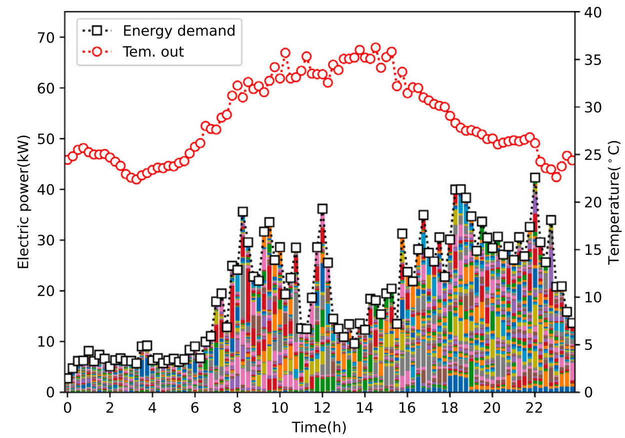


Fig. 3. The outdoor temperatures and energy demand of each prosumer.

4.3. Problem reformulation

To evaluate the worst-case expectation from the problem (9), assume that a series of samples are sampled from the support. The deviation of samples $(\hat{\xi}_i^0, \hat{\xi}_i^1, \dots, \hat{\xi}_i^{m+1})$ is arranged in ascending order, where $\hat{\xi}_i^0$ and $\hat{\xi}_i^{m+1}$ are the minimal and maximal deviations within the support, respectively. Considering that the probability distribution of forecasted PV power generation is consistent with the forecasted deviation. The worst-case expectation is rewritten as follows.

$$\sup_{\mathbb{P}_i \in \mathcal{P}_i} \int_{\Xi_i} C_i(\boldsymbol{x}_i, \xi_i)\mathbb{P}(d\xi_i) \tag{24a}$$

$$s.t.\quad \underline{\mathrm{p}}_i^k \le \int_{\Xi_i} \mathbb{I}_{[\hat{\xi}_i^0, \hat{\xi}_i^k]}(\xi_i)\mathbb{P}(d\xi_i) \le \overline{\mathrm{p}}_i^k \quad k = 1, 2, \dots, m+1 \tag{24b}$$

where $\mathbb{I}_{[\hat{\xi}_i^0, \hat{\xi}_i^k]}(\xi_i)$ is an indicator function. It is 1 when $\xi_i \in [\hat{\xi}_i^0, \hat{\xi}_i^k]$ otherwise is 0.

Assigning dual variables $(\overline{\lambda}_k, \underline{\lambda}_k)$ to constraints (24)b, the dual problem is given as follows.

$$\inf_{\overline{\lambda}_k, \underline{\lambda}_k} \sum_{k=1}^{m+1} \overline{\lambda}_k \overline{\mathrm{p}}_i^k - \underline{\lambda}_k \underline{\mathrm{p}}_i^k \tag{25a}$$

$$s.t.\quad \sum_{k=1}^{m+1} (\overline{\lambda}_k - \underline{\lambda}_k)\mathbb{I}_{[\hat{\xi}_i^0, \hat{\xi}_i^k]}(\xi_i) - C_i(\boldsymbol{x}_i, \xi_i) \ge 0 \tag{25b}$$

$$\overline{\lambda}_k, \underline{\lambda}_k \ge 0 \qquad k = 1, 2, \dots, m+1 \tag{25c}$$

It is observed that $\xi_i \in [\hat{\xi}_i^0, \hat{\xi}_i^{m+1}]$ can be partitioned into $m+1$ mutually disjoint $[\hat{\xi}_i^{k-1}, \hat{\xi}_i^k]$. Furthermore, since the affine adjustment ensures a convex function over a compact interval, the minimization/maximization is attained on the boundaries $[\hat{\xi}_i^{k-1}, \hat{\xi}_i^k]$. Therefore, the constraint (25)b can be rewritten as follows.

$$\sum_{k=j}^{m+1} (\overline{\lambda}_j - \underline{\lambda}_j) - C_i(x, \hat{\xi}_i^{k-1}) \ge 0 \tag{26a}$$

$$\sum_{k=j}^{m+1} (\overline{\lambda}_j - \underline{\lambda}_j) - C_i(x, \hat{\xi}_i^k) \ge 0 \tag{26b}$$

The original infinite-dimensional optimization problem (9) is converted to a deterministic linear optimization problem and thus can be further rewritten as (27). The derived LP counterparts model can be solved by Gurobi, CPLEX, MOSEK solver, etc.

$$\min\quad C_i^{p2p} + \sum_{k=1}^{m+1} \overline{\lambda}_k \overline{\mathrm{p}}_i^k - \underline{\lambda}_k \underline{\mathrm{p}}_i^k \tag{27a}$$

$$s.t.\quad (9)b\text{–}(9)f, (25)c, (26)a, (26)b \tag{27b}$$

Remark 3. The counterparts model will vary with different ambiguity sets and the derived LP counterparts model is tailor-made. It is worth mentioning that the counterpart model from the prior DRO-based P2P energy trading [23] is not applicable to our work.

5. Numerical results

In this section, we provide numerical results for the performance evaluation of the proposed decentralized approach. The formulation and method are programmed in Python and the solver is Gurobi 9.5.1 which runs on a Windows 10 PC with 64 GB of RAM, 28 logical cores, and a speed of 3.3 GHz.

5.1. Basic data

The operation of the proposed P2P energy trading was verified using the IEEE European Low Voltage Test Feeder [41]. The maximal power import/export is set as ±14 kW. Each prosumer has their own HVAC unit and ESS unit. The parameters of the HVAC unit is set as C_i = 3 kWh/°C, R_i = 8 °C/kW, η_i = 3, and maximum energy consumption is 3.5 kW for all prosumers from [42]. The indoor temperature ranges from 22 °C to 25 °C. The energy storage level is limited in [1, 10] kWh, the maximum charging and discharging power is both 2 kW, and the charging and discharge efficiency is 0.94 and 1.06. The purchasing power prices offered by the utility are flat prices set as 0.25 \$/kWh, and the feed-in prices are 0.05 \$/kWh. The levelized cost of ESS is 0.02 \$/kWh. The bilateral trade weight is 3\$/(kWh · distance unit). The parameter of the ADMM is set as $\psi = 2$.

The solar dataset includes the environmental and electrical data [43], which is located in Brazil (latitude −20.510886 longitude −54.619878) and the time range is from Oct. 2019 to Mar. 2020. Considering that the cost of PV and ESS have declined over the past decade, prosumers will have a high willingness to install them in the future. However, some special geographical factors and inconvenient housing conditions limit the ability of some prosumers to install PV, while this limitation is inapplicable to ESS because it is a small-footprint device. It is assumed that 11 prosumers with rooftop PV systems, of which the maximum power is 8.3 kW. The total number of prosumers is 55 within the energy community and all of them are equipped with ESS. The outdoor temperatures and energy demand of each prosumer are shown in Fig. 3.

Table 1
The absolute value of correlation coefficients between PV power and environmental attributes.

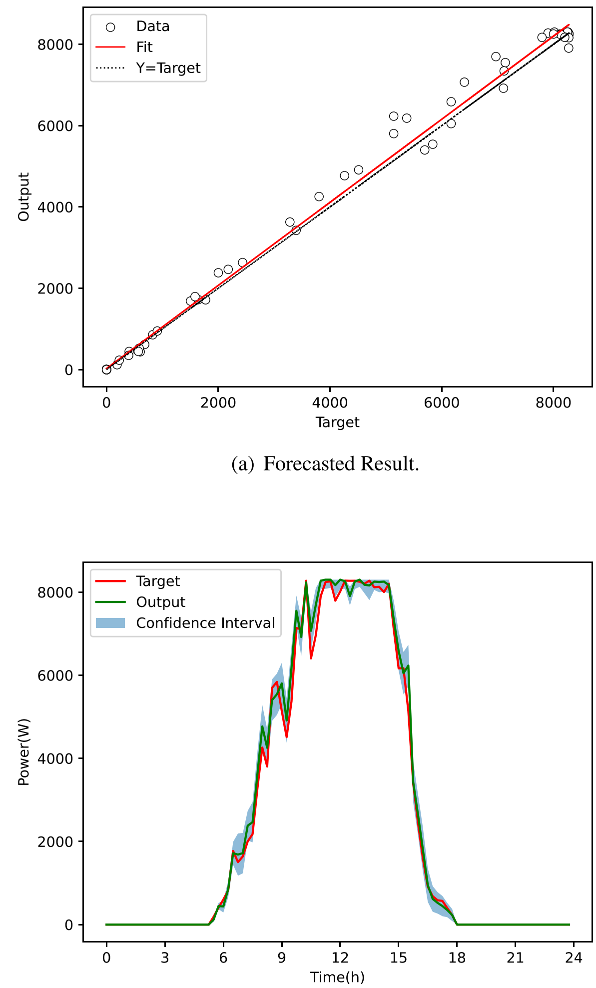
Environmental attributes	Pearson coefficient
solar irradiation	**0.9927**
environment temperature	**0.7623**
average particulate concentration	**0.4024**
1 μm particulate mass	**0.1383**
1 μm particulate concentration	0.1382
2 μm particulate mass	0.1335
2 μm particulate concentration	0.1207
rain precipitation	0.1095
10 μm particulate concentration	0.0238
4 μm particulate concentration	0.0227
10 μm particulate mass	0.0176
4 μm particulate mass	0.0173

5.2. Forecast results and analysis

The dataset from [43] consists of 12 environmental attributes, including wind speed, wind direction, temperature, solar irradiation, pluviometer, mass, and concentration of particulate, while the forecasted attribute is generated PV generation. For mining the correlations between the environmental attributes and solar power, the Personal correlation coefficient is used and the result is shown in Table 1.

Based on the evaluation result, we choose four most relevant attributes, i.e., solar irradiation, environment temperature, average particulate concentration, and 1 μm particulate mass, to forecast solar power generation. Then, a sub-dataset is extracted from the overall data, which consists of 96 time slots per day for 15 days. The sub-dataset is split into two parts, one is the training set (14 days) and the other is the testing set (1 day). The day-ahead PV generation forecast is realized by bagging DGP. The number of weak learners and the percentage of subsampling are set as 10 and 90% by trial and error, respectively. The forecasted PV generation result and interval are shown in Fig. 4.

The results of performance evaluation on the same data set are shown in Table 2. The hyperparameters of the BDL-based algorithm [28] are summarized as follows. The activation function is selected as ReLu and the learning rate is set as 0.01, the network layers of BDL are set as 4, and neurons in the hidden layers are chosen to be 512 and 256. The kernel function of the GP-based algorithm is selected as the radial basis function. The number of inducing points for the DGP-based algorithm is 128. The number of weak learners is set as 10 for bagging BDL and bagging GP. For further evaluating the performance of the forecast algorithm, the non-parametric Holm–Bonferroni procedure [44] in relation to the PV generation forecasted problem is presented in this paper. The results of the ranking are presented in Table 3, which clearly indicates that the bagging DGP achieved the highest rank among the other algorithms. Specifically, while the results of the Holm–Bonferroni procedure indicated that the bagging DGP did not significantly outperform the bagging BDL, bagging GP, and GP, it clearly did significantly outperform the other competitors. In the view of time complexity analysis, the time complexity of the BDL-based algorithm is $\mathcal{O}(NK^2)$, where K is the number of neurons within the BDL. The time complexity of the GP-based algorithm is $\mathcal{O}(N^3)$. Since the DGP-based algorithm is a stacked sparse GP, the time complexity is also $\mathcal{O}(NM^2)$. It is indicated that the order of time complexity is GP > BDL ≈ DGP. It is worth pointing out that the uncertainty-aware prosumer prefers a forecast result with higher trueness (i.e., the closeness of the mean value to the true value) and lower standard deviation (i.e., the level of spread of the values in the forecast result). Therefore, the BDL-based approach is not preferred by the prosumers due to the Pinball loss. Furthermore, the vast quantity of information brought by EMS is a challenge to GP in view of time feasibility. High-performance hardware-supported EMS is not always affordable for the prosumers within the smart energy community. Therefore, the proposed DGP is a good choice for prosumers considering both the precision and hardware aspects.



(a) Forecasted Result.

(b) Forecasted interval.

Fig. 4. Day-ahead PV Power Forecast Result and 95% confidence interval.

Table 2
Performance evaluation on day-ahead power forecast.

Approach	ICP	Pinball	R^2
BDL	94.8%	155.7	0.98763
GP	86.5%	42.7	0.99631
DGP	84.4%	47.2	0.99447
Bagging BDL	**100.0%**	168.5	0.99636
Bagging GP	87.5%	42.4	0.99631
Bagging DGP	96.9%	**35.7**	**0.99643**

5.3. Simulation result

The mean value of the day-ahead P2P energy trading schedule for peers within the energy community is shown in Fig. 5. Here, the negative values indicate that the energy is sold to other peers or the utility. During the time 8:00–16:00 when PV generation is high, the P2P energy trading between peers is sufficient to cover local loads, charge the battery, and feedback to the utility. The prosumer with rooftop PV systems prefers to sell surplus energy to other prosumers or charge the battery, rather than fed it back to the utility. The prosumer prefers to use their local energy storage system in the early morning and night to reduce operating costs during P2P energy trading. It should be noted that some prosumers prefer to feed back to utility due to the exogenous cost within the P2P energy trading. Furthermore, the total cost is cut off by about 48.22% when compared with no P2P energy trading. The

Table 3
Holm–Bonferroni procedure in terms of error indices; reference algorithm = bagging DGP (Rank = 5.67).

j	Algorithms	Rank	z_j	p_j	δ/j	Hypothesis
1	Bagging BDL	4	−1.290	9.84e−2	5.00e−2	Accept
2	Bagging GP	3.67	−1.549	6.07e−2	2.50e−2	Accept
3	GP	3.33	−1.807	3.54e−2	1.67e−2	Accept
4	BDL	2.33	−2.582	0.49e−2	1.24e−2	Rejected
5	DGP	2.00	−2.840	0.23e−2	1.00e−2	Rejected

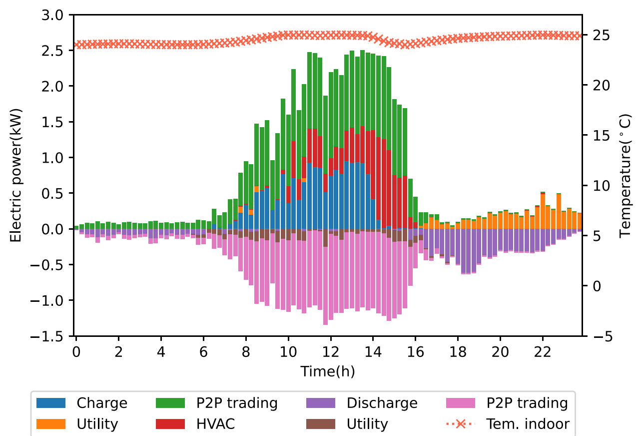


Fig. 5. The mean value of P2P energy trading simulation result for peers.

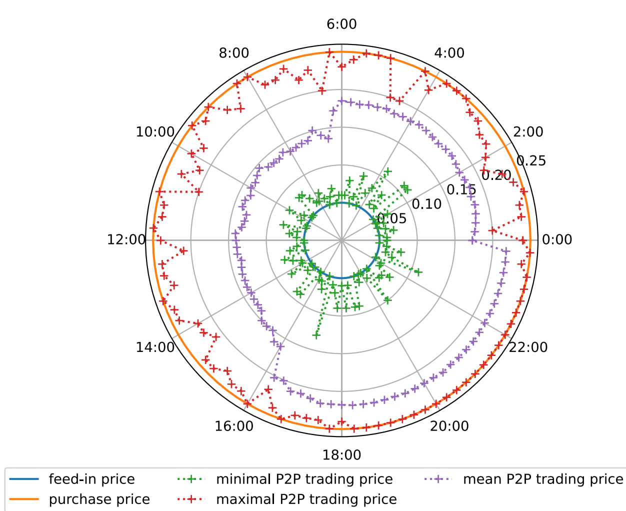


Fig. 6. The price for P2P energy trading.

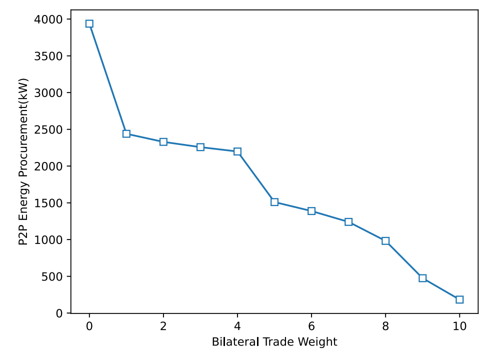


Fig. 7. The profile of P2P energy procurement and bilateral trade weight.

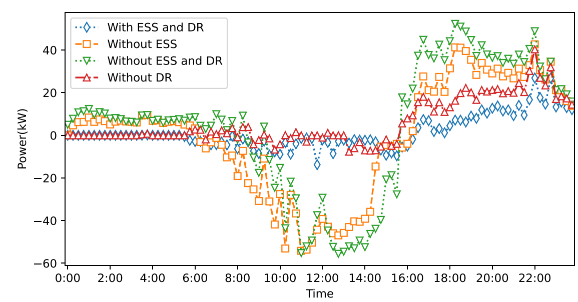


Fig. 8. The volume of power imported/exported from the utility.

result highlights the economical advantages of the proposed P2P energy trading scheme.

The price of P2P energy trading, which is derived through the dual variable ($\boldsymbol{\pi}$) from the ADMM algorithm is shown in Fig. 6. The result demonstrates that the P2P price is within the interval between the purchasing price and feed-in price, and thus it increases the enthusiasm of prosumers to participate in P2P energy trading. Moreover, the low P2P energy trading price is encountered when the local energy power is sufficient, and high when the local energy is scarce. It is worth noting that most prosumers prefer to trade with peers through local ESS in the early morning and charge in the noon when local solar energy is abundant.

The profile of bilateral trade weight and P2P energy procurement is shown in Fig. 7. It is shown that the volume of P2P energy trading is inversely correlated with the bilateral trade weight. It can be anticipated that when bilateral trade weight is high-enough, the prosumer will not favor P2P energy trade but trade with the utility.

5.4. Comparison of results

In this subsection, we have compared the result with different prosumer preferences. Four preferences are considered in this paper and shown as follows: (1) P2P energy trading without ESS; (2) P2P energy trading without DR; (3) P2P energy trading without both ESS and DR. and (4) P2P energy trading with both ESS and DR (baseline approach). The temperature is set as 23.5 °C for the without DR mechanism case, which is the mean value within the temperature range. The power imported/exported from the utility is shown in Fig. 8. The energy import profile without ESS case presents a dual-peak characteristic, i.e., the power is fed back to the grid during the day while a demand peak occurs at night. Compared with the ESS-equipped scenarios, energy trading is cut down significantly. Similarly, the volume of power imported/exported is also reduced with the application of the DR mechanism.

The total cost of prosumers with different preferences is shown in Table 4. It can be observed that the application of ESS and DR are the preferred options for prosumers compared to other competing schemes. Moreover, compared to the baseline with ESS and DR, the cost will increase by 23.56% if no ESS is equipped, by 42.34% if no DR is deployed, and by 67.24% if neither ESS nor DR is deployed. These results highlight the economic potential of ESS and DR in P2P energy trading.

Table 4
The cost of prosumers with different preferences.

Case	Cost($)
Without ESS and DR	101.58
Without DR	86.46
Without ESS	75.05
With ESS and DR (Baseline)	**60.74**

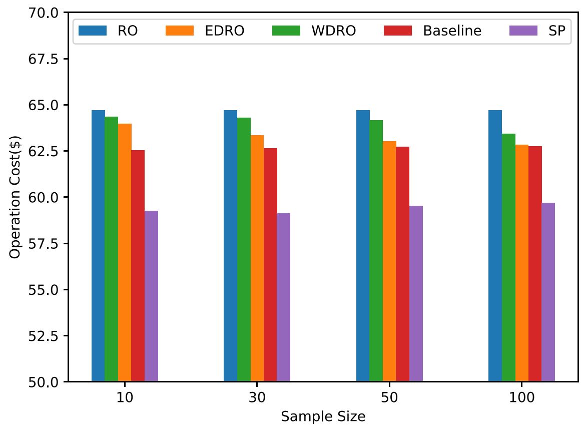


Fig. 9. Optimal cost comparison of different optimal approaches under different sample sizes.

5.5. Performance evaluation

The performance of our proposed DRO is compared with four benchmark approaches, i.e., Wasserstein-based DRO (WDRO) [27], Empirical cumulative distribution-based DRO (EDRO) [26], RO and SP [24]. The radius of the Wasserstein ball refers to *Proposition 1* in [45]. The result of day-ahead PV generation is obtained by bagging implicit posterior variational inference DGP. The solution quality is demonstrated by the sampling size in the proposed model, specifically, and the sampling size is 10, 30, 50, and 100 within the support, respectively.

Fig. 9 shows the optimal value of the total operation cost within one day for all 55 prosumers. It is worth pointing out that the optimal operation cost is based on the result of the day-ahead PV forecast, i.e., the corresponding cost is the in-sample cost. The result demonstrates that the RO is the most conservative approach as it has the highest operating cost. Conversely, the SP yields the lowest cost due to it underestimates the worst-case scenario. For the DRO-based approach, it provides a conservative in-sample cost considering that the ambiguity set of WDRO and EDRO is formed by a statistical approach. In contrast, the proposed DRO is based on a data-driven ambiguity set, which describes the uncertainty more precisely and performs a low in-sample cost. It is worth pointing out that when the sampling size within the ambiguity set is large, the DGP estimated probability distribution can also be well-fitted by the empirical cumulative distribution. However, our proposed DRO is more sample-efficient compared with EDRO.

The total social cost in real-time is shown in Table 5, the energy imbalance in the real-time stage is realized by trading with the utility. It should be noted that the real PV generation has both samples within the forecasted interval and outside the interval (shown in Fig. 4b). It is observed that the proposed approach has the most modest social cost when compared with other uncertainty optimization methods. Furthermore, due to the forecasted result of bagging DGP overestimated the PV power (shown in Fig. 4a), which causes a slightly unrealistic sample interval. RO mainly concentrates on the worst-case scenario, that is, the minimum PV power expected output. The out-of-sample is closer to the worst-case scenario and thus RO results in a less social cost when compared with SP. The DRO-based approach considers worst-case expectation, which overcomes the over-conservativeness of the worst-case scenario considered in RO, thus resulting in even lower operating costs in the real-time stage. It is worth noting that although the differences between the proposed approach and other DRO-based approaches are not significant. When the community has more energy demand, especially for the business and industrial prosumers, insignificant gaps can lead to differences in the energy production costs for a large-scale community.

Table 5
The real-time cost of different approaches.

Optimization approach	Cost($)
Baseline (This paper)	**62.378**
EDRO	62.628
WDRO	63.297
RO	64.412
SP	66.657

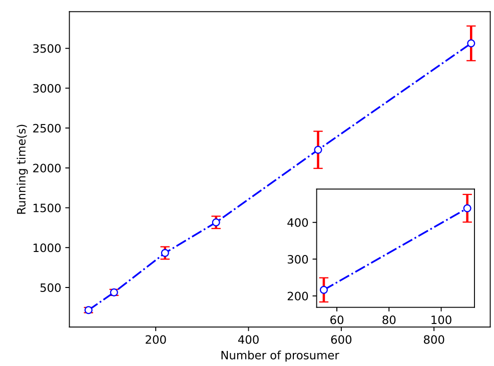


Fig. 10. Impact of prosumers on computation time.

5.6. Scalability

For testing the scalability of the proposed approach, a large number of prosumers are employed in P2P energy trading. The IEEE European Low Voltage Test Feeder System is augmented by adding new prosumers. The prosumers with rooftop solar power panels account for 20% of the total prosumers. The average running time and feasible running time range (calculated over 10 runs) for varying numbers of prosumers are shown in Fig. 10. In the case of more prosumers, the computation time is increased when compared with fewer prosumers cases. But its growth rate is approximately linear with the increase of prosumer size, and hence it is feasible in scalability.

It is worth mentioning that the delay of signal is not considered in the above simulation result. In real-life application projects, communication latency and channel congestion are critical issues that limit the scalability of P2P energy trading. In specific scenarios, some sparse techniques can be adapted to resolve the communication burden. By contrast, centralized and/or distributed approaches require more hardware resources in the coordinator node as the scale of prosumers increases. The reliability of the centralized/distributed approach is also questionable when the coordinator is at fault.

6. Concluding remarks

In this paper, we proposed a DRO-based optimal model for P2P energy trading. The proposed approach is a fully data-driven DRO model, especially in the ambiguity set construction with the aid of bagging DGP. In the proposed model, the expected operating costs and operational constraints of each prosumer are formed in a DRO problem. The worst-case distribution in the ambiguity set is considered and achieves a robust operation for P2P energy trading. The numerical result indicated the economic perspective of P2P energy trading. Specifically, both the simulation results of P2P energy trading and the application of ESS and DR have good economic performance. Furthermore, the conservatism of in-sample costs is shrunk when compared with other RO-based and DRO-based competitors, and real-time costs are modest among competitors. Moreover, this is the first attempt to integrate DRO and ML into P2P energy trading.

Further improvements on this work will take two specific directions. The first will involve the network constraints in the P2P energy trading, while the second will involve the trusted intermediary with blockchain technology.

CRediT authorship contribution statement

Xihai Zhang: Conceptualization, Methodology, Software, Writing – original draft, Investigation, Validation, Writing – review & editing. **Shaoyun Ge:** Validation, Supervision, Writing – review & editing. **Hong Liu:** Methodology, Funding acquisition, Supervision, Writing – review & editing. **Yue Zhou:** Validation, Writing – review & editing. **Xingtang He:** Validation, Writing – review & editing. **Zhengyang Xu:** Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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